

Dynamical Systems in Biology

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Abstract

Dynamical systems theory provides a useful framework for analyzing complex biological signals that exhibit irregular, yet potentially structured behavior. This paper surveys two representative applications: using chaos theory to control cardiac arrhythmias and Lyapunov-exponent-based analysis of electroencephalogram (EEG) signals in schizophrenia. In the former, experimentally measured interbeat intervals are shown to admit a low-dimensional return-map description. This enables the stabilization of unstable periodic orbits using minimal feedback. In the latter, Lyapunov exponents extracted from EEG time series are used as quantitative features reflecting local dynamical stability, with attention to estimator dependence. These case studies are complemented by brief examples across biology, illustrating how concepts such as instability, invariant structure, and bifurcation recur across biological scales without requiring fully detailed mechanistic models.

1 Introduction

Biological systems frequently generate time series data that are irregular, high-dimensional, and strongly nonlinear. Traditional linear or statistical approaches often treat such variability as noise, obscuring underlying structure. Over the past several decades, chaos theory and nonlinear dynamical systems have offered an alternative perspective that irregular biological signals may arise from deterministic dynamics organized around instabilities, invariant sets, and bifurcations.

This assignment summarizes two concrete realizations of this perspective. The first is the experimental control of cardiac arrhythmias using chaos-control principles, and the second is the analysis of resting-state EEG signals in schizophrenia using Lyapunov exponents. Although these applications differ in physiological system, data modality, and clinical objective, they share a common methodological core: inference about stability properties from reduced, experimentally accessible observables.

To place these two case studies in a broader context, the paper also briefly surveys selected applications of dynamical systems theory across biology, including population ecology, systems biology, neuroscience, and renal physiology. These examples highlight that the same mathematical frameworks can be appropriated across diverse biological domains, reinforcing the generality of the dynamical-systems approach. To facilitate accessibility, this work also includes a brief appendix section with definitions for key terms used throughout the paper.

The goal of this paper is to clarify how abstract concepts from nonlinear dynamics are translated into biologically-relevant insights. Rather than emphasizing full mechanistic modeling, the focus is on how dynamical structure can be inferred, interpreted, and, in some cases, manipulated directly from data.

2 Experimental and Mathematical Analysis of Chaos Control in Cardiac Arrhythmia

This section presents a detailed experimental and mathematical description of the landmark study *Controlling Cardiac Chaos* by Garfinkel, Spano, Ditto, and Weiss (1992). The work is notable for demonstrating, in a real cardiac preparation, that irregular arrhythmic dynamics can be interpreted as deterministic chaos and stabilized using small, state-dependent perturbations derived from nonlinear dynamical systems theory.

2.1 Background: Cardiac Dynamics and Nonlinear Systems

This section introduces the minimal biological and mathematical concepts required to interpret cardiac arrhythmias using dynamical systems theory. Only those elements directly used in subsequent analysis are discussed.

2.2 Cardiac excitation as a nonlinear oscillator

Cardiac muscle cells are electrically excitable and generate action potentials through nonlinear ionic currents. At the single-cell level, the transmembrane voltage $V(t)$ is commonly modeled by

$$C_m \frac{dV}{dt} = -I_{\text{ion}}(V, \mathbf{w}) + I_{\text{stim}}(t),$$

with auxiliary recovery variables $\mathbf{w}$. Under periodic stimulation, this system exhibits stable oscillations corresponding to regular heartbeats.

A key physiological property is refractoriness: after each action potential, a finite recovery time is required before the cell can be re-excited. This introduces 'memory' and nonlinearity into beat-to-beat dynamics.

2.3 Propagation and reentry in cardiac tissue

In tissue, cells are electrically coupled, and excitation propagates as waves. A standard continuum description is a reaction–diffusion system

$$\frac{\partial V}{\partial t} = D \nabla^2 V - \frac{1}{C_m} I_{\text{ion}}(V, \mathbf{w}),$$

which supports traveling and rotating waves. Under pathological conditions, waves may circulate repeatedly (reentry), forming spiral waves in two dimensions and scroll waves in three dimensions. Reentry provides a dynamical mechanism for sustained arrhythmias.

2.4 Arrhythmias as dynamical phenomena

From a dynamical systems perspective it is established that a normal rhythm corresponds to a stable periodic orbit, simple arrhythmias correspond to higher-period or quasiperiodic oscillations, and fibrillation corresponds to complex, often aperiodic spatiotemporal dynamics. The central question this paper is trying to answer is whether irregular rhythms are stochastic or arise from deterministic chaos.

2.5 Experimental observables

In experiments, the full cardiac state is inaccessible. Instead, electrical activation times are recorded at one or more locations. Let t_n denote successive activation times; the primary 'observable' is the interbeat interval

$$I_n = t_{n+1} - t_n.$$

The sequence $\{I_n\}$ serves as a reduced representation of the underlying dynamics.

2.6 Return maps and discrete-time dynamics

Beat-to-beat behavior is analyzed using a return (or Poincare) map

$$x_n = (I_{n-1}, I_n), \quad x_{n+1} = F(x_n).$$

Periodic rhythms appear as fixed or finite sets of points, while chaotic rhythms generate extended invariant sets. Unstable fixed points of this map correspond to unstable periodic cardiac rhythms embedded within irregular activity.

2.7 Rationale for chaos control

Electrical stimulation can reliably shorten interbeat intervals by triggering early activation. If arrhythmic dynamics are deterministic near unstable periodic orbits, then small, state-dependent timing perturbations can stabilize these orbits by steering trajectories toward their stable manifolds.

This motivates chaos-control approaches based on local linearization of experimentally constructed return maps, as developed in the following section.

2.8 Experimental preparation and measured signals

The experimental system consisted of an isolated, perfused rabbit interventricular septum. The tissue was perfused through the septal branch of the left coronary artery with oxygenated Krebs solution maintained at 37°C. Electrical activity was recorded using surface Ag–AgCl electrodes that measured monophasic action potentials (MAPs). Electrical stimulation was delivered via platinum electrodes using a constant-voltage pulse of duration approximately 3 ms and amplitude in the range 10–30 V.

Arrhythmic activity was induced pharmacologically by adding a chemical called ouabain (2–5 μM), often in combination with epinephrine (2–10 μM), to the perfusate. Under these conditions, the preparation typically transitioned from near-periodic beating to complex, highly irregular rhythms resembling ventricular tachyarrhythmia.

The MAP signal was digitized in real time at a sampling rate of 2 kHz with 12-bit resolution. Individual beats were detected algorithmically as the time of the maximum upstroke velocity, i.e.,

$$t_n = \arg \max \left(\frac{dV}{dt} \right),$$

where $V(t)$ denotes the recorded voltage signal. From this, the primary time series analyzed was the sequence of interbeat intervals

$$I_n = t_{n+1} - t_n.$$

This scalar sequence served as the sole observable for both chaos diagnosis and control.

2.9 Return map construction and identification of chaos

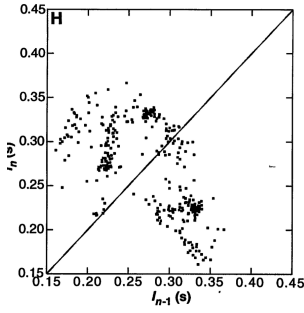
To analyze the dynamics, the authors constructed a first-return (Poincare) map from the interbeat intervals:

$$x_n = (I_{n-1}, I_n) \in R^2,$$

which implicitly defines a discrete-time dynamical system

$$x_{n+1} = F(x_n).$$

Periodic rhythms correspond to a finite number of points on this map: a single point for period-1, two points for period-2, and so forth. In contrast, during the irregular arrhythmic state, the return map exhibited a structured but non-periodic distribution of points, inconsistent with pure noise and suggestive of deterministic chaos. Both are illustrated in the following figure:



Subfigure A and B demonstrate the MAP at a fixed point and it's corresponding return map. Subfigure C and D demonstrate the MAP demonstrating period-2 orbit and it's corresponding return map. Subfigure H shows the return map of an aperiodic MAP Source: *Controlling Cardiac Chaos* (1992)

A key observation was the presence of an unstable fixed point

$$x_F = (I^*, I^*)$$

lying near the diagonal $I_n = I_{n-1}$. Short sequences of successive beats showed trajectories that approached this point along one direction and then departed along another. This geometry is characteristic of a *flip saddle*, a fixed point with one stable and one unstable direction, where iterates alternate sides of the stable manifold. Such behavior is a classical signature of low-dimensional chaotic dynamics.

2.10 Local linearization and dynamical interpretation

Near the fixed point x_F , the return map can be locally approximated by a linear system

$$x_{n+1} - x_F \approx A(x_n - x_F),$$

where $A = DF(x_F)$ is the Jacobian matrix. Let λ_u denote the unstable eigenvalue of A with $|\lambda_u| > 1$, and let $\vec{f}_u$ be the corresponding unstable eigenvector. Components of perturbations along $\vec{f}_u$ grow exponentially under iteration, while components along the stable eigenvector decay.

Experimentally, λ_u was estimated by observing the rate at which successive points diverged from the fixed point along the unstable direction:

$$\lambda_u \approx \frac{\langle x_{n+1} - x_F, \vec{f}_u \rangle}{\langle x_n - x_F, \vec{f}_u \rangle},$$

for points sufficiently close to x_F . Importantly, this estimation relied entirely on measured data and did not require a biophysical model of ionic currents.

2.11 Control objective and physiological constraints

The control objective was to stabilize the unstable periodic orbit associated with x_F using minimal intervention. Physiologically, the only reliable control input was the ability to *advance* the timing of the next beat by applying an electrical stimulus before the next spontaneous activation. Let p denote the predicted time of the next natural beat; then a control action consisted of delivering a stimulus at time $p - \delta p$, thereby shortening the current interbeat interval.

A crucial constraint is that the experimenters could reliably shorten intervals but could not lengthen them in a controlled manner, i.e., $\delta p \geq 0$. This unidirectional control restriction strongly influenced the form of the control algorithm and the types of periodic rhythms that could be stabilized.

2.12 Proportional Perturbation Feedback (PPF)

The authors implemented a control strategy termed *Proportional Perturbation Feedback* (PPF), conceptually similar to the Ott–Grebogi–Yorke (OGY) method for parameter control. However, this mechanism was adapted to direct state perturbations rather than OGY’s parameter modulation.

Near the fixed point, the controlled dynamics may be written as

$$x_{n+1} - x_F \approx A(x_n - x_F) + b u_n,$$

where $u_n = \delta p$ is the timing advance and b encodes the sensitivity of the return map to this perturbation. The control law was chosen as

$$\delta p = C (x_n - x_F) \cdot \vec{f}_u,$$

where C is a proportionality constant selected to counteract growth along the unstable direction.

The constant C depends on both the unstable eigenvalue λ_u and a sensitivity factor g , defined experimentally as

$$g \approx \frac{\delta x_F}{\delta p},$$

which measures how the location of the fixed point shifts in response to a small timing advance. These quantities were estimated during an initial observation phase lasting between 5 and 60 seconds.

2.13 Two Phases of Control

The control protocol consisted of two distinct phases:

1. **Learning phase:** The system evolved freely while the algorithm constructed the return map, identified the fixed point x_F , estimated stable and unstable directions, and determined the sensitivity

parameter g .

2. **Intervention phase:** Whenever the trajectory entered a small neighborhood of x_F , the control law was activated. A stimulus was applied only when needed, producing a corrected iterate closer to the stable manifold.

Geometrically, each control step nudged the system state toward the stable manifold of the unstable periodic orbit, preventing escape along the unstable direction. Repeated small corrections resulted in long-term stabilization of a low-period rhythm, most commonly of period approximately 3.

2.14 Validation and experimental outcomes

Control was attempted in 11 experimental runs and succeeded in 8. Successful control was characterized by sustained periodic behavior with only intermittent stimulation. When control was turned off, the system reliably returned to chaotic behavior, often passing through transient complex periodic states. Furthermore, fixed-rate pacing at the same mean frequency as the control interventions failed to stabilize the rhythm, demonstrating that stabilization relied on state-dependent feedback rather than average rate effects.

An additional observation was that chaos control often reduced the mean heart rate, despite individual control actions shortening specific intervals. This was attributed to the suppression of extreme short intervals that normally followed very long ones in the flip-saddle dynamics.

An example of a successfully controlled case is presented here:

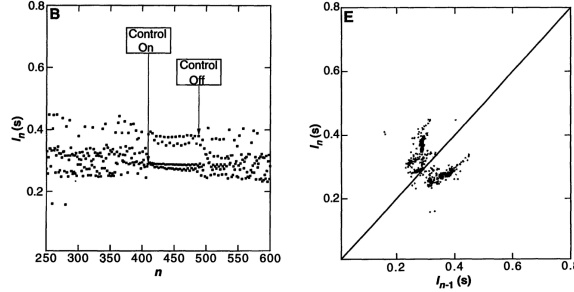


Figure 1: Source: *Controlling Cardiac Chaos* (1992). Explanation below.

Subfigure (B) plots the interbeat interval I_n as a function of beat number n . Prior to control activation, I_n exhibits large, irregular fluctuations, consistent with aperiodic dynamics. At the indicated time ($n \approx 400$), feedback control is switched on. Immediately thereafter, the variability collapses and I_n converges to a small set of repeating values, indicating stabilization of a low-period rhythm. When control is switched off, the system rapidly returns to irregular behavior, demonstrating that the periodic rhythm is maintained by feedback rather than intrinsic pacing.

Figure (E) shows the corresponding Poincaré return map (I_{n-1}, I_n) . Instead of a single stable fixed point, the data form multiple localized clusters away from the diagonal $I_n = I_{n-1}$. These clusters correspond to a stabilized periodic orbit of finite period (typically period ≈ 3), rather than a period-1 rhythm. The diagonal line is shown as a reference; points do not collapse onto it, confirming that control does not enforce simple fixed-rate pacing.

The geometry of the return map is consistent with control of an unstable periodic orbit embedded in a chaotic attractor. Feedback perturbations act locally to suppress divergence along the unstable manifold of the orbit, keeping trajectories confined near its stable manifold. The coexistence of multiple clusters reflects the discrete nature of the stabilized orbit in state space and provides direct geometric evidence that chaos control is achieved through invariant-manifold stabilization rather than global rhythm suppression.

2.15 Conceptual significance

This experiment represents a concrete realization of chaos control in biology. It demonstrates that cardiac arrhythmias can exhibit low-dimensional deterministic chaos and that unstable periodic orbits embedded in chaotic dynamics can be identified directly from data. We also see that small, precisely timed perturbations are sufficient to stabilize these orbits, and that control operates through geometric manipulation of invariant manifolds. The work provides a foundational application of chaos theory to experimentally realizable cardiac electrophysiology.

2.16 Subsequent Developments in Mathematical Control of Cardiac Arrhythmias

After the initial demonstration that irregular cardiac rhythms could be stabilized by targeting unstable periodic orbits embedded in chaotic dynamics [6], the field converged on the view that arrhythmias arise from identifiable dynamical instabilities in excitable media rather than generic low-dimensional chaos. Beat-to-beat 'alternans' was reinterpreted as a period-doubling instability of an underlying return map, while fibrillation was framed as spatiotemporal chaos driven by spiral-wave and filament dynamics. This shifted control objectives from rhythm suppression on a global scale to local modification of stability properties [11].

3 Lyapunov-Exponent Features in Schizophrenia EEG

3.1 Motivation

A recurring hypothesis in nonlinear EEG analysis is that schizophrenia may alter the *local stability* of cortical dynamics. It was established that these cortical dynamics can be summarized (imperfectly) by Lyapunov exponents computed from resting-state EEG. Kutevov et al., in their work *EEG analysis in patients with schizophrenia based on Lyapunov exponents* (2020), characterise an aspect of this. Instead of treating the Lyapunov exponent as a single agreed-upon measurement, they compare classical time-series estimators for the *largest* Lyapunov exponent (LLE) and also compute a short Lyapunov *spectrum* as a multi-feature alternative [12].

3.2 Key terms and minimal setup

Each EEG electrode is treated as a scalar time series

$$x(t_k), \quad k = 1, \dots, N,$$

with sampling rate $f_s = 128$ Hz and 60 s duration (so $N = 7680$ samples per channel), analyzed channel-wise [12].

To interpret a scalar signal as a dynamical system trajectory, the standard step is delay embedding:

$$\mathbf{x}_i = (x_i, x_{i-\tau}, x_{i-2\tau}, \dots, x_{i-(m-1)\tau}) \in R^m,$$

with delay τ (in samples) and embedding dimension m [17]. Kutepov et al. fix $m = 4$ and report a nominal “5 ms” delay choice. Implementation details aside, the operative object is an integer lag consistent with 128 Hz sampling [12].

3.3 What was known

Largest Lyapunov exponents from data are *estimator-dependent* in noisy and finite time-series recordings such as EEG. Classical estimators differ in various ways like how neighborhoods are formed in reconstructed space and how temporal correlations are handled. The paper’s review explicitly shows that earlier schizophrenia EEG Lyapunov findings as inconsistent enough to motivate an algorithm comparison rather than assuming a single canonical pipeline [12].

3.4 What the paper does

Data Two cohorts of adolescents are analyzed: schizophrenia ($n = 45$, ages 10–14) and controls ($n = 39$), with 16 EEG channels (10–20 system), resting eyes-closed; artifacts are removed manually, and each channel is processed independently [12].

Core computation: estimate instability from neighbor divergence. They compute LLE using three standard time-series estimators: Wolf, Rosenstein, and Kantz [20, 15, 10, 12].

I just present the operational formulas in this section rather than re-deriving them for brevity:

1. **Wolf et al.’s method** – Track the growth of separations between a reference trajectory and a nearby neighbor, periodically replacing the neighbor when the separation exceeds a threshold; estimate

$$\hat{\lambda}_1 \approx \frac{\sum_{k=1}^M \ln(\varepsilon'_k / \varepsilon_k)}{\sum_{k=1}^M T_k \Delta t},$$

where ε_k and ε'_k are the initial and terminal separations over segment k , and T_k is the segment length [20].

2. **Rosenstein et al.’s method: mean log-divergence slope** – For each embedded point $\mathbf{x}_i$, choose a nearest neighbor $\mathbf{x}_{i'}$ (with temporal separation constraints), define

$$d_i(k) = \|\mathbf{x}_{i+k} - \mathbf{x}_{i'+k}\|, \quad \left\langle \ln d_i(k) \right\rangle \approx \text{const} + \lambda_1(k \Delta t),$$

and estimate $\hat{\lambda}_1$ as the slope over an identified linear regime [15].

3. **Kantz’s method: neighborhood-averaged divergence** – For each reference $\mathbf{x}_t$, form a neighborhood

$$U_t = \{\mathbf{x}_i : \|\mathbf{x}_i - \mathbf{x}_t\| \leq r\},$$

compute the stretching function

$$S(\tau) = \frac{1}{T} \sum_{t=1}^T \ln \left(\frac{1}{|U_t|} \sum_{i \in U_t} \|\mathbf{x}_{t+\tau} - \mathbf{x}_{i+\tau}\| \right),$$

and use $\hat{\lambda}_1 = \text{slope of } S(\tau) \text{ versus } \tau \text{ over a linear regime [10]}$.

Spectrum features (beyond λ_1). They also compute a short Lyapunov spectrum (here, consistent with $m = 4$) using a Jacobian-like approach attributed to Sano–Sawada / related methods: locally fit a linear map A_j that advances neighbor difference vectors,

$$\mathbf{y}_{i+m} \approx A_j \mathbf{y}_i, \quad A_j = \arg \min_A \frac{1}{N} \sum_i \|\mathbf{y}_{i+m} - A \mathbf{y}_i\|^2,$$

then accumulate average log-growth to obtain $(\lambda_1, \dots, \lambda_m)$ [16, 12].

Group comparison For each algorithm and each channel, they compute per-subject Lyapunov features and run standard group comparisons (reported as channel-wise averages and p -values), treating channels with $p \leq 0.05$ as significant discriminators [12].

3.5 Results

Their core empirical conclusion is that *estimator choice changes which channels appear informative*. Specifically, they report that Wolf- and Kantz-based LLE features are not very representative for their classification framing, whereas Rosenstein’s method is the most representative and yields six relevant channels [12].

They further report that using a Lyapunov *spectrum* increases the number of significant channels relative to using only λ_1 , consistent with the idea that discriminative information may sit in multiple expanding/contracting directions of the reconstructed dynamics, not only the leading one [12].

4 Selected Applications of Dynamical Systems in Biology

4.1 Population Ecology

Population ecology models species abundances as state variables evolving under nonlinear ordinary differential equations. A canonical single-species model is logistic growth,

$$\dot{x} = rx \left(1 - \frac{x}{K} \right),$$

introduced historically in the context of population growth limits by Verhulst [18].

Interacting populations lead to coupled nonlinear systems, such as the predator–prey (Lotka–Volterra) model,

$$\dot{x} = x(\alpha - \beta y), \quad \dot{y} = y(-\gamma + \delta x),$$

which was developed in early mathematical biology and ecology by Lotka and Volterra [13, 19]. Ecological “regime shifts” and resilience are often framed as changes in attractor structure and basin geometry;

the modern ecological resilience concept is classically articulated by Holling [7].

4.2 Specific Cases in Systems Biology

Systems biology models biochemical networks as nonlinear dynamical systems

$$\dot{x} = f(x, p),$$

where x denotes molecular concentrations and p kinetic parameters. Positive feedback can generate multistability (multiple stable equilibria separated by saddles), which has been demonstrated in synthetic gene circuits such as the genetic toggle switch [5].

Negative feedback can generate oscillations (limit cycles) under appropriate nonlinearities and timescales; a canonical synthetic example is the repressilator [2]. Robust biological regulation is often cast in control-theoretic terms. For example, network structure enabling robust adaptation without fine-tuning was analyzed in the context of bacterial chemotaxis by Barkai and Leibler [1].

4.3 Neuroscience

Single neurons are modeled as excitable dynamical systems with fast–slow structure,

$$\dot{V} = f(V, w), \quad \dot{w} = g(V, w),$$

where V is membrane voltage and w a recovery variable. The FitzHugh model provides a canonical planar reduction capturing excitability and recovery [4], and the Nagumo–Arimoto–Yoshizawa circuit provides a related traveling-pulse realization [14].

Transitions between rest and repetitive firing are organized by bifurcations (e.g. Hopf and SNIC), and the general nonlinear-dynamics toolkit for spiking and coupled neural systems is treated systematically by Ermentrout and Terman [3].

4.4 Nephron Coupling

Nephrons can exhibit oscillatory dynamics due to a phenomena called tubuloglomerular feedback (TGF), and interactions between nephrons can produce synchronization phenomena. Empirical evidence for nephron–nephron interaction via synchronization of intratubular pressure oscillations was reported by Holstein-Rathlou [8]. A dynamical model of the TGF mechanism (supporting oscillatory behavior and phase relationships) was developed by Holstein-Rathlou and Marsh [9].

A minimal abstraction treats nephrons as coupled oscillators,

$$\dot{x}_i = f(x_i) + \sum_j \kappa_{ij}(x_j - x_i),$$

where x_i represents pressure/flow in nephron i and κ_{ij} represents coupling through shared vasculature; collective dynamics then include phase-locking, partial synchronization, and wave-like coordination.

5 Conclusion

The examples discussed in this paper illustrate how nonlinear dynamical systems theory can be applied productively to biological systems. In the cardiac case, irregular arrhythmic dynamics are shown to organize around unstable periodic orbits that can be identified directly from data and stabilized using small, state-dependent perturbations. This demonstrates that effective intervention need not suppress dynamics globally, but can instead target specific instability mechanisms embedded in the system. In the neural case, Lyapunov-exponent-based analysis of EEG highlights a complementary use of dynamical concepts. Here, instability measures serve as descriptive features rather than direct control targets, and their interpretation depends strongly on estimator choice and data limitations. This study contributes by highlighting the importance of methodological caution when applying invariant-based quantities to noisy, high-dimensional biological signals. The additional examples from ecology, systems biology, neuroscience, and nephron dynamics emphasize that such issues are not domain-specific. Taken together, these studies support the view that chaos theory has genuine explanatory value in biology, particularly as a tool for understanding instability, structure, and control in complex dynamical systems.

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Appendix: Essential Vocabulary

This appendix defines a *minimal* set of biological and dynamical-systems terms used throughout the paper. The aim is to facilitate readers from either background to parse the arguments without external references.

Biological definitions

Time series A sequence of measurements indexed by time or beat number (e.g. voltage, interbeat intervals).

Action potential A transient, nonlinear spike in membrane voltage generated by excitable cells (cardiac or neuronal).

Excitable system A system that responds to sufficiently strong input with a large, stereotyped event followed by recovery.

Refractory period A post-excitation interval during which the system cannot (or can weakly) respond to new stimuli.

Activation time A time marker assigned to each excitation, typically defined as the moment of fastest voltage upstroke,

$$t_n = \arg \max \left(\frac{dV}{dt} \right),$$

used to discretize continuous signals into beat events.

Interbeat interval (IBI) The time between successive activations,

$$I_n = t_{n+1} - t_n,$$

commonly used as a reduced observable of cardiac dynamics.

Arrhythmia Any deviation from normal, regular cardiac rhythm, arising from abnormal impulse generation, propagation, or both.

Alternans Beat-to-beat alternation in electrical or calcium properties (long–short–long–...), often preceding arrhythmia.

Reentry A self-sustaining loop of electrical activation in tissue, producing rapid or irregular rhythms.

Fibrillation A severe arrhythmia characterized by disorganized, high-frequency activation with loss of coherent wavefronts (atrial or ventricular), corresponding to spatiotemporally complex dynamics.

Spiral wave A rotating wave of excitation in excitable media; its core organizes reentrant and fibrillatory dynamics.

Physiological noise Variability from measurement, cellular heterogeneity, or unresolved degrees of freedom, superimposed on deterministic dynamics.

Ouabain / epinephrine (arrhythmia induction) Pharmacologic agents used experimentally to increase propensity for irregular rhythms.

EEG channel / electrode A scalp recording site that yields a voltage time series reflecting aggregate postsynaptic activity; channels are often analyzed separately or jointly.

Sampling rate The measurement frequency f_s (Hz) of a discretized signal; sets time step $\Delta t = 1/f_s$ and constrains resolvable dynamics.

Cohort / control group Groups of subjects used for comparison (e.g. clinical vs. healthy controls), typically analyzed via per-subject feature extraction then group-level statistics.

Tubuloglomerular feedback (TGF) A renal regulatory loop where tubular salt delivery influences afferent arteriolar tone and thus filtration; it can generate intrinsic oscillations and inter-nephron synchronization.

Mathematical definitions

Dynamical system A rule that evolves a state in time: $\dot{x} = f(x)$ (continuous), $x_{n+1} = F(x_n)$ (discrete).

State space The space of all possible system states; trajectories are curves (or sequences) within this space.

Fixed point A state that reproduces itself:

$$f(x^*) = 0 \quad \text{or} \quad F(x^*) = x^*.$$

Represents steady behavior in the reduced description.

Periodic orbit A repeating trajectory with period T (or period p for maps), corresponding to regular rhythms.

Stability Whether small perturbations decay (stable) or grow (unstable) under the dynamics.

Return map A discrete map constructed by sampling a continuous system once per cycle (e.g. beat-to-beat dynamics).

Invariant set / attractor A set of states preserved by the dynamics that captures long-term behavior.

Chaos Deterministic dynamics that are aperiodic and sensitive to initial conditions.

Lyapunov exponent A quantitative measure of average exponential divergence of nearby trajectories; positive values indicate instability.

Bifurcation A qualitative change in system behavior caused by variation of a parameter (e.g. onset of alternans).

Local linearization Approximation of dynamics near a state by its first derivative, used to analyze stability and control.